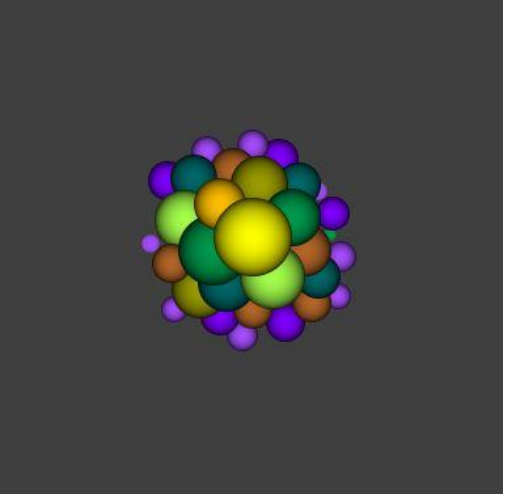
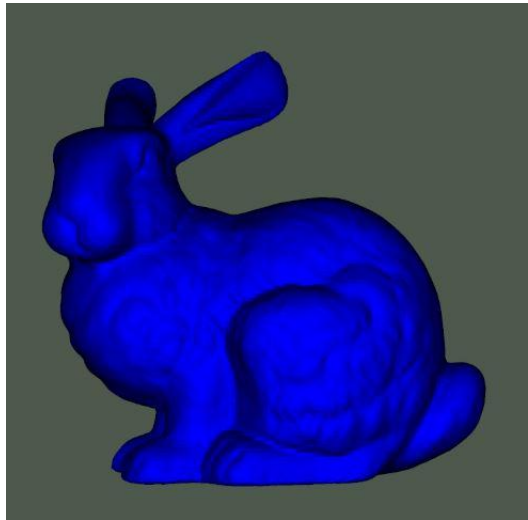


CS 440: Computer Graphics

Week 5

Lecture 10: Affine Transformations and Mesh Representations



Acknowledgements: Cem Yuksel, University of Utah; Waqar Saleem, Habib University

Announcements and Recap

- Assignment due Friday (please follow submission instructions)
- Quiz Results out

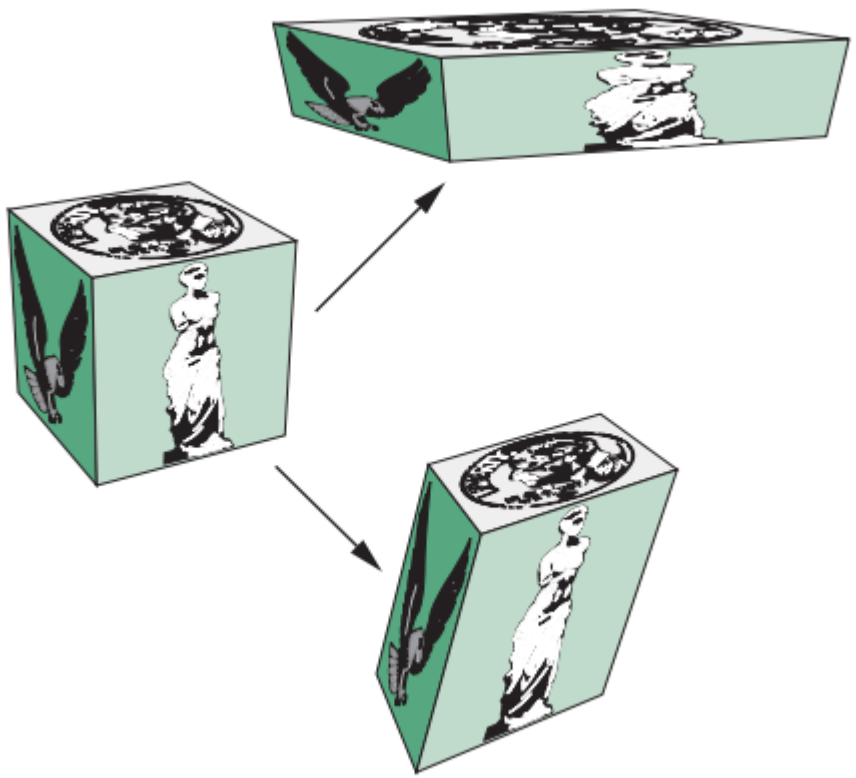
- Recap:
  - Affine Transformations
    - Transformation Matrix
    - Translation
    - Rotation
    - Scaling
    - Shear

Today: Affine Transformations and Mesh Representations

- Concatenating Transformations
- General Transformations
- Model Instance Paradigm
- Scene Graphs
- 3D Geometry
- Topology
- Mesh Data Structures
- Mesh Analysis

Rigid vs Non-Rigid Body Transformations

- Is the shape or volume of an object preserved after transformation?
- Examples?
  - Non-rigid body: Shear, Scaling
  - Rigid body: Translation, Rotation

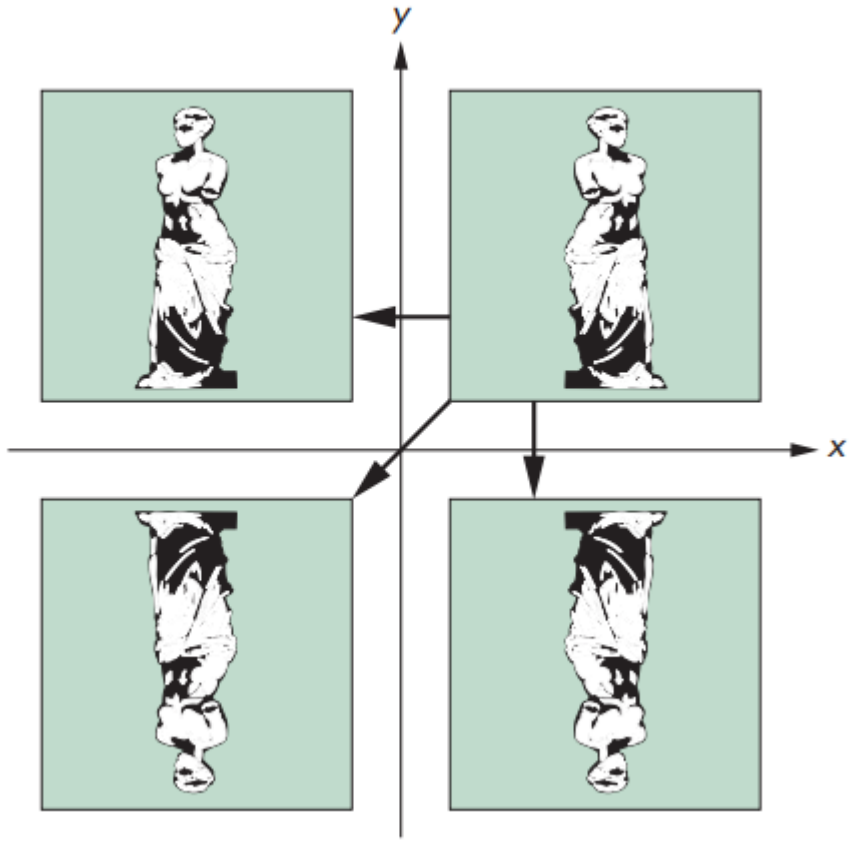


Some Previously Unanswered Questions…

- What happens when we apply scaling to points?

- Reflections!

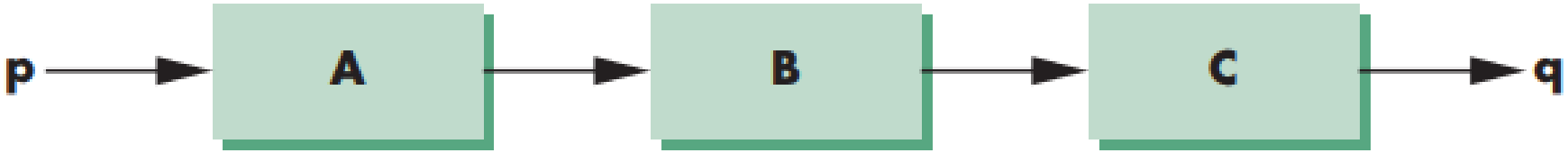
$$R_{xy} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



Concatenating Transformations

$$\mathbf{q} = CBA\mathbf{p}$$

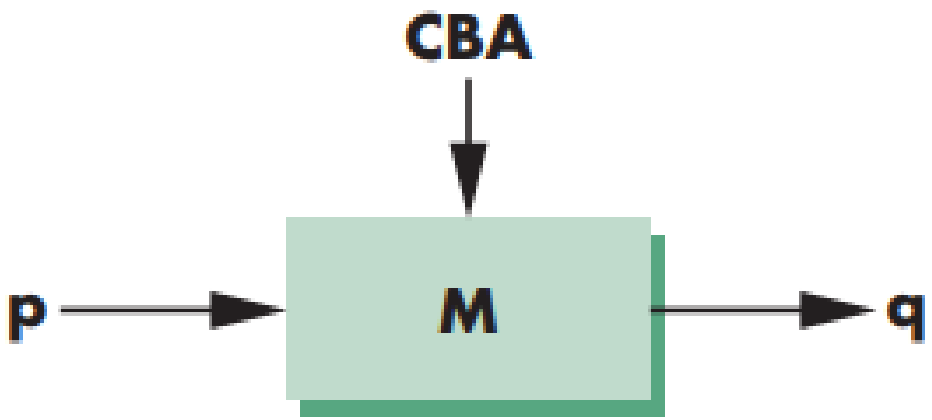
$$\mathbf{q} = (C(B(A\mathbf{p})))$$



Concatenating Transformations

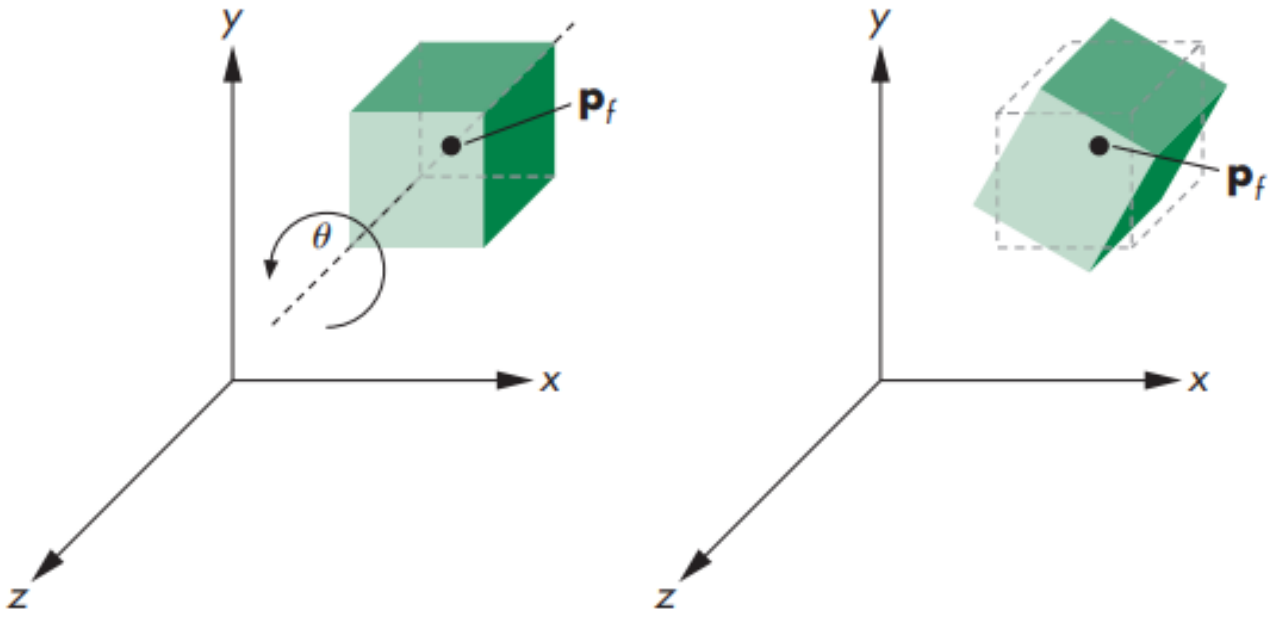
$$\mathbf{q} = \mathbf{M}\mathbf{p}$$

$$\mathbf{M} = CBA$$



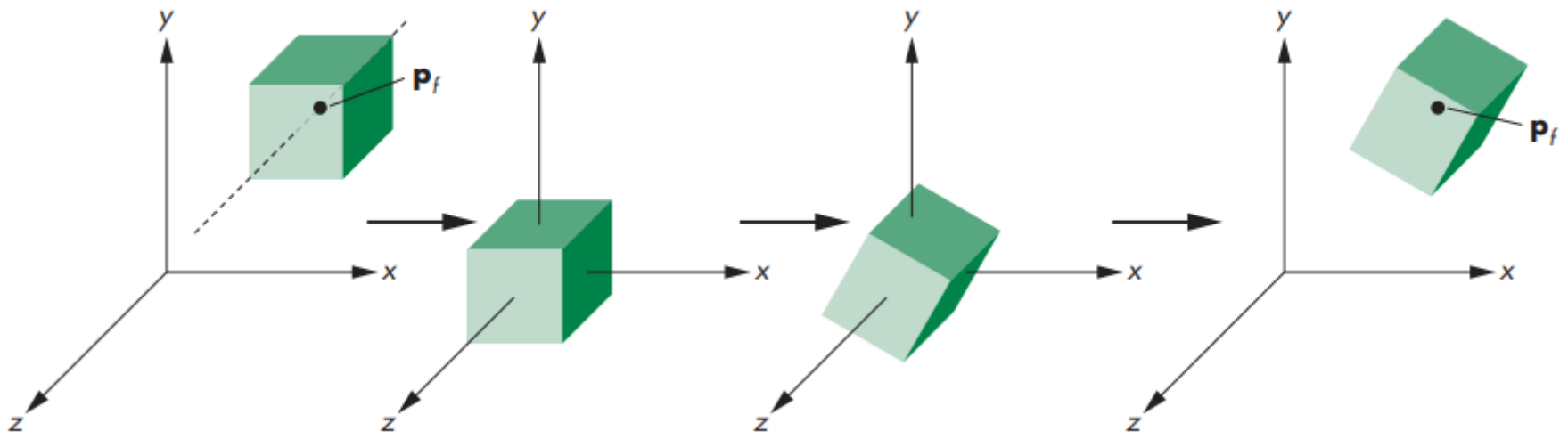
Rotation: Changing the Fixed Point

- How to rotate the cube around its center?



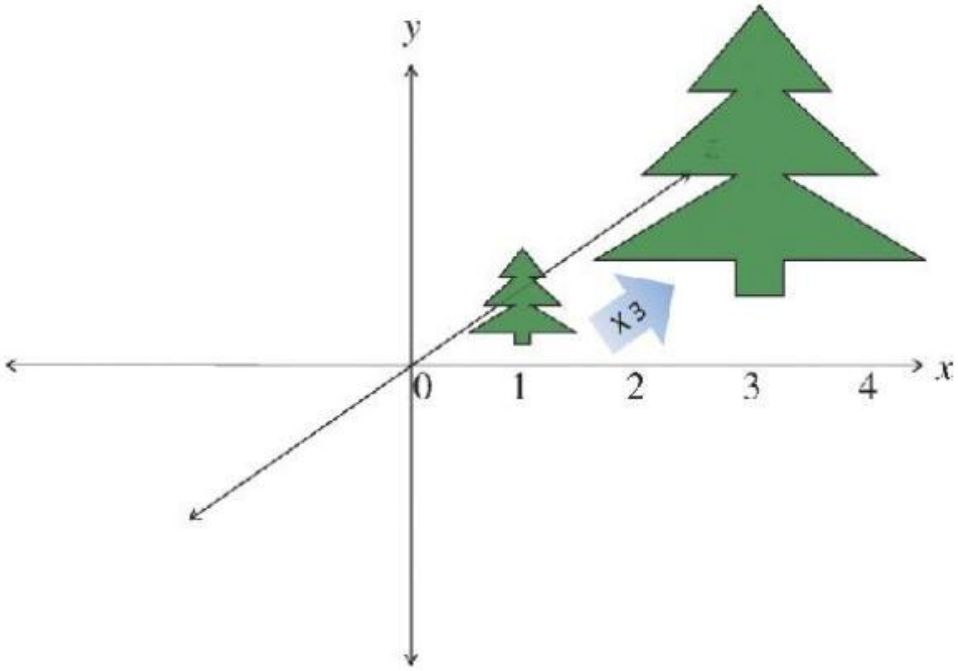
Rotation: Changing the Fixed Point

- Translate to Origin
- Rotate along Z-axis
- Translate back to original position

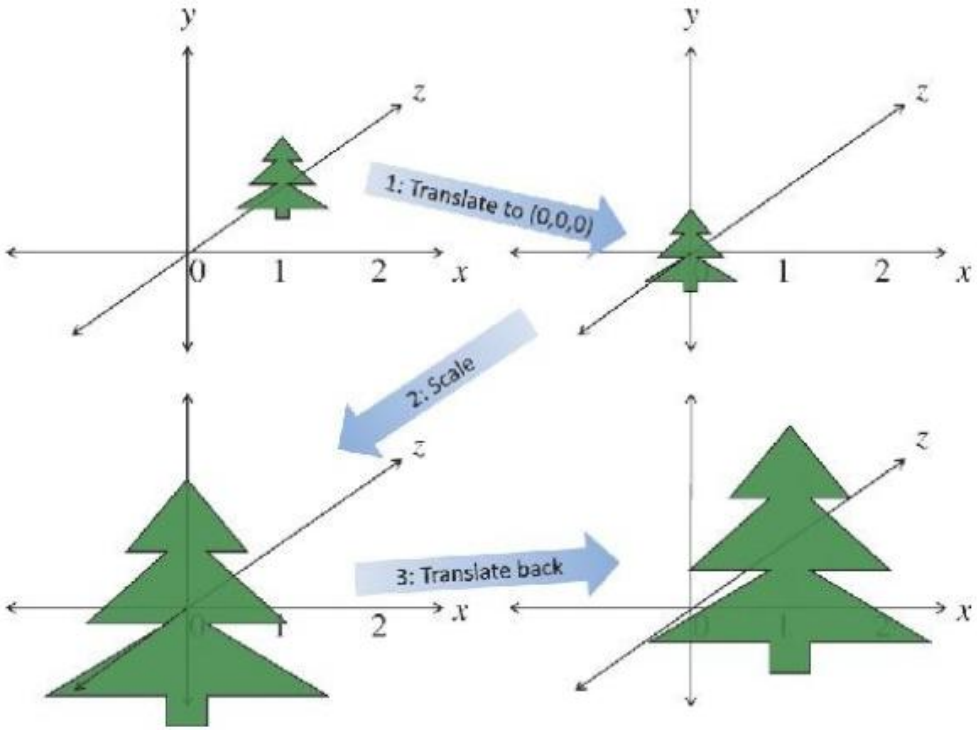


Scaling: Changing the Fixed Point

Scaling w.r.t Origin



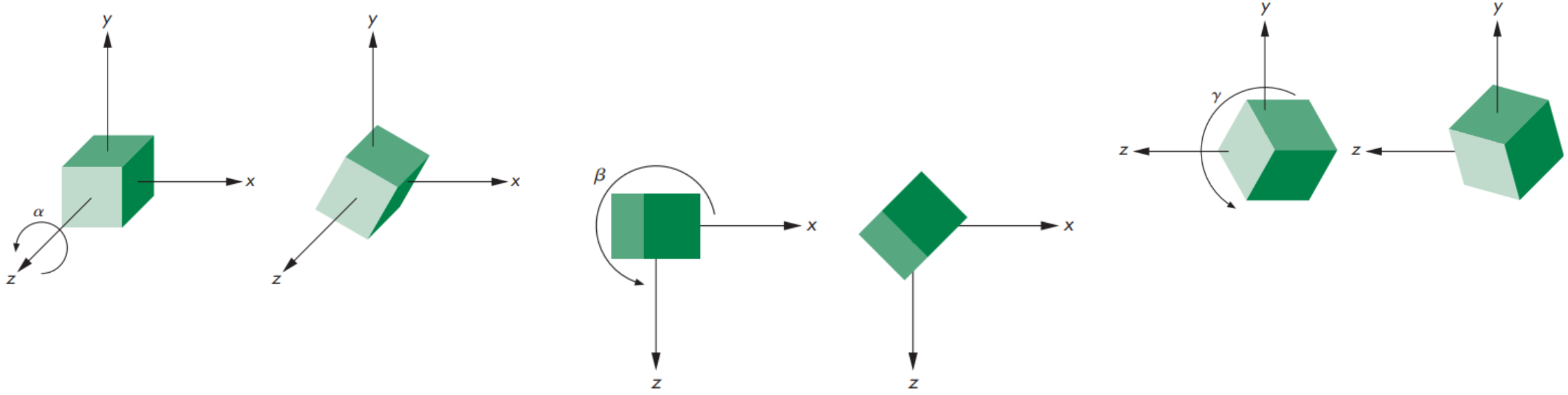
Scaling w.r.t a Fixed Point



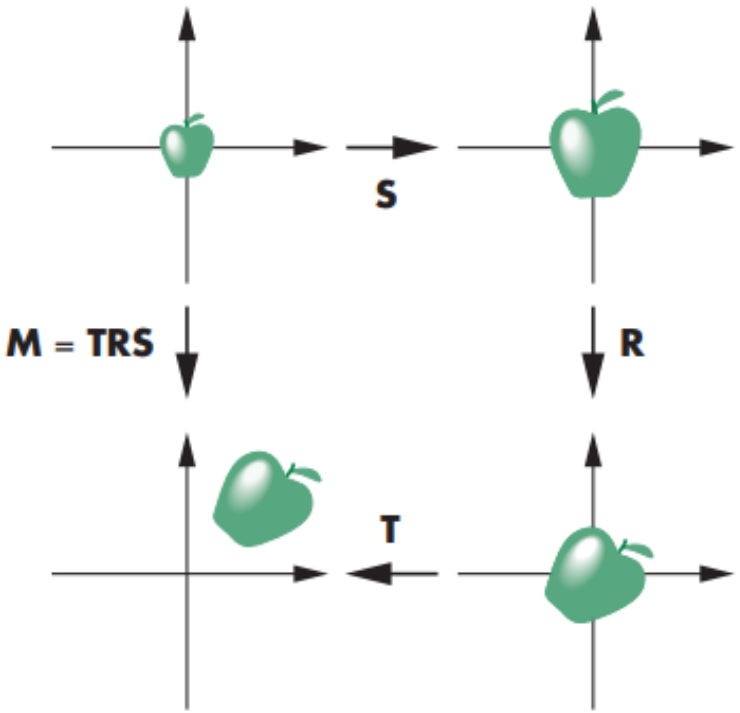
General Rotation

- Order of transformation

$$R = R_x R_y R_z$$

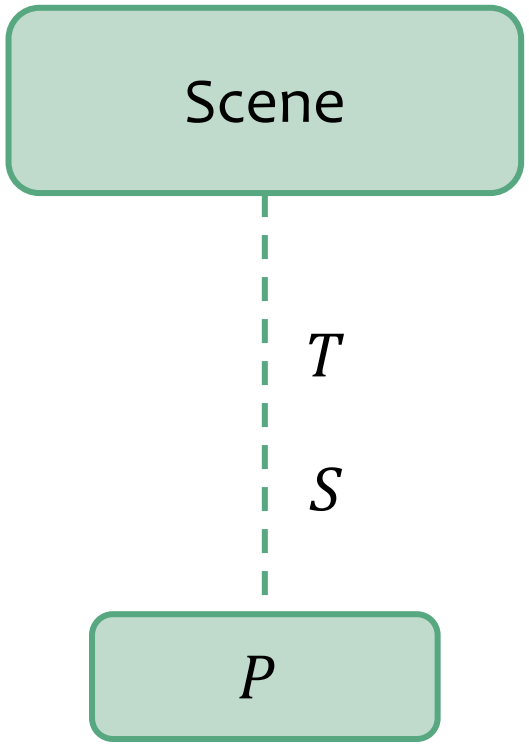


Instance Transformation



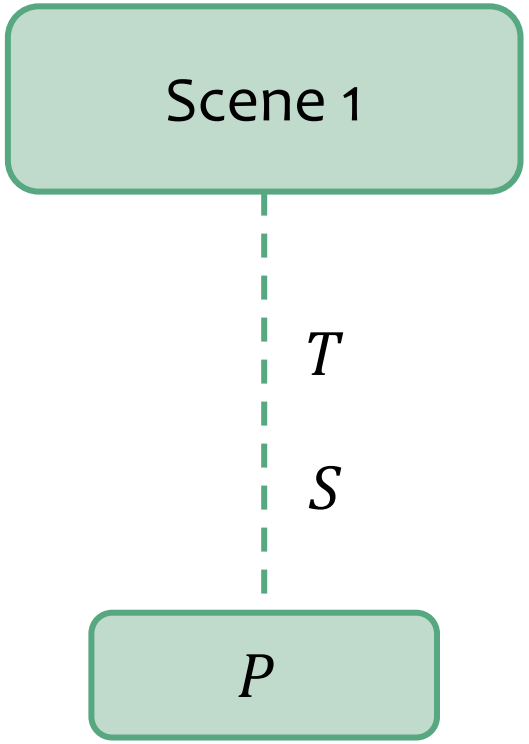
Scene Graphs

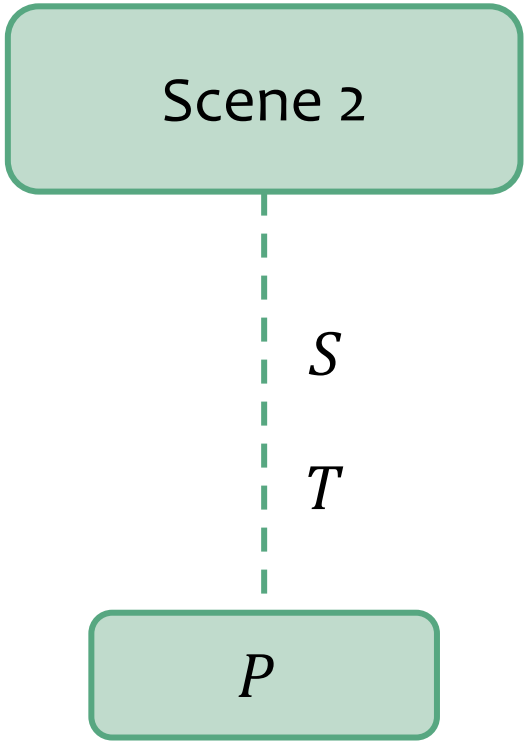
- Complex scenes may have a lot of objects
  - 3D models, triangles, spheres…
- A lot of transformations!
- How do we organize them all?



$$P' = T \cdot S \cdot P$$

Scene Graphs

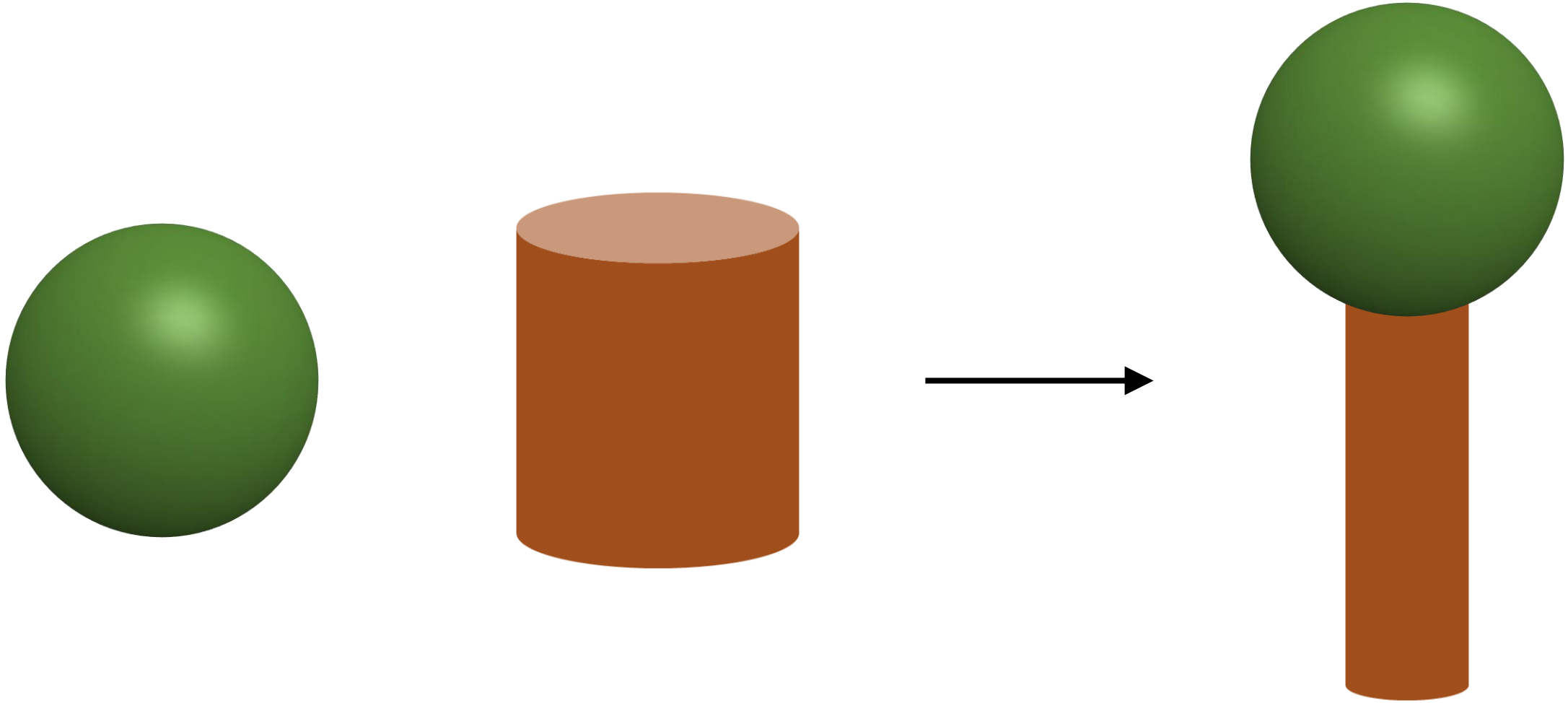




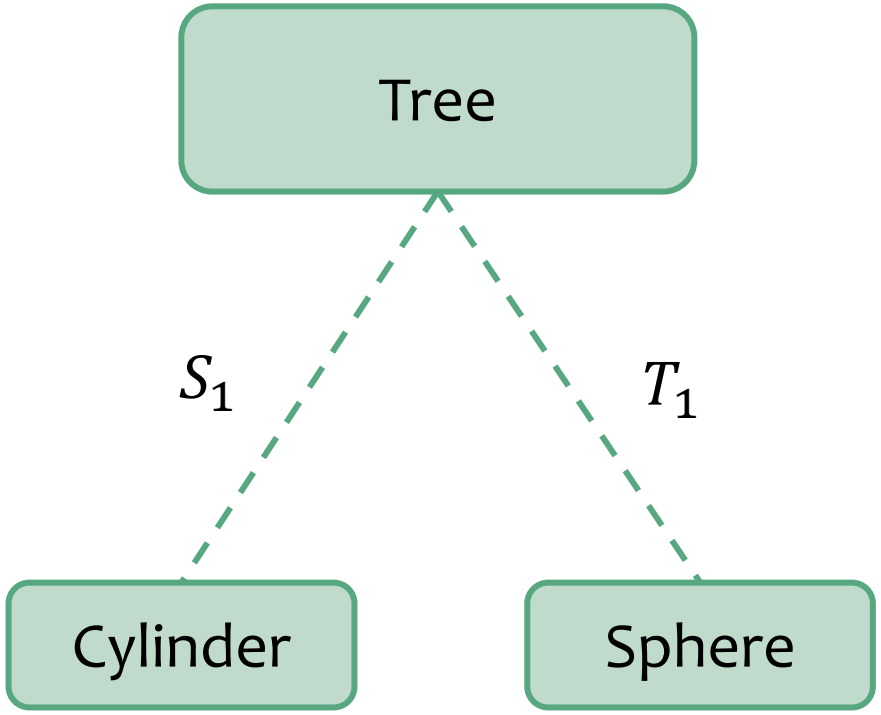
$$P' = T \cdot S \cdot P$$

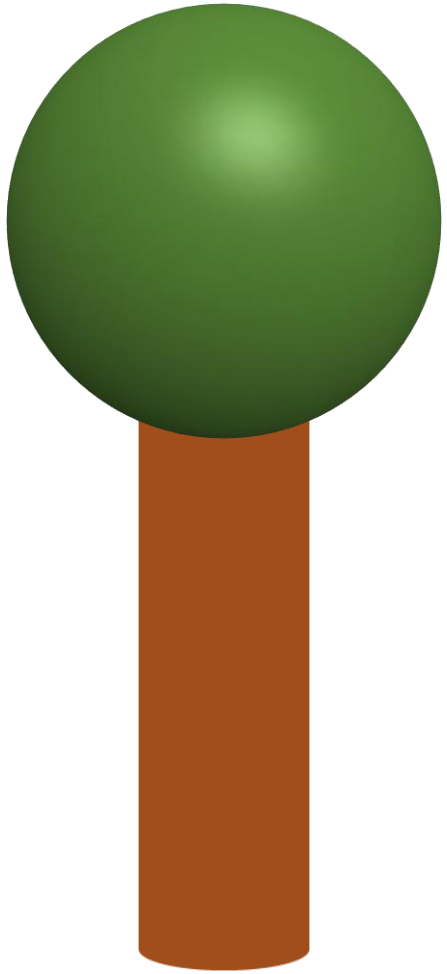
$$P'' = S \cdot T \cdot P$$

Scene Graphs



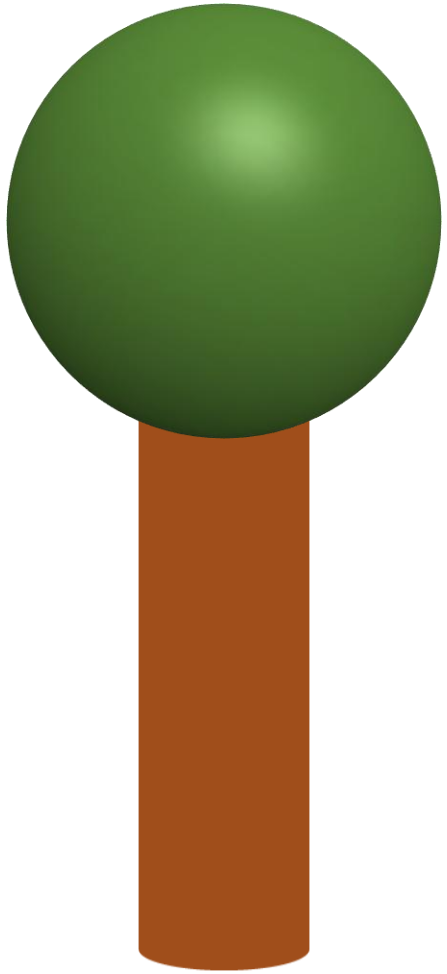
Scene Graphs



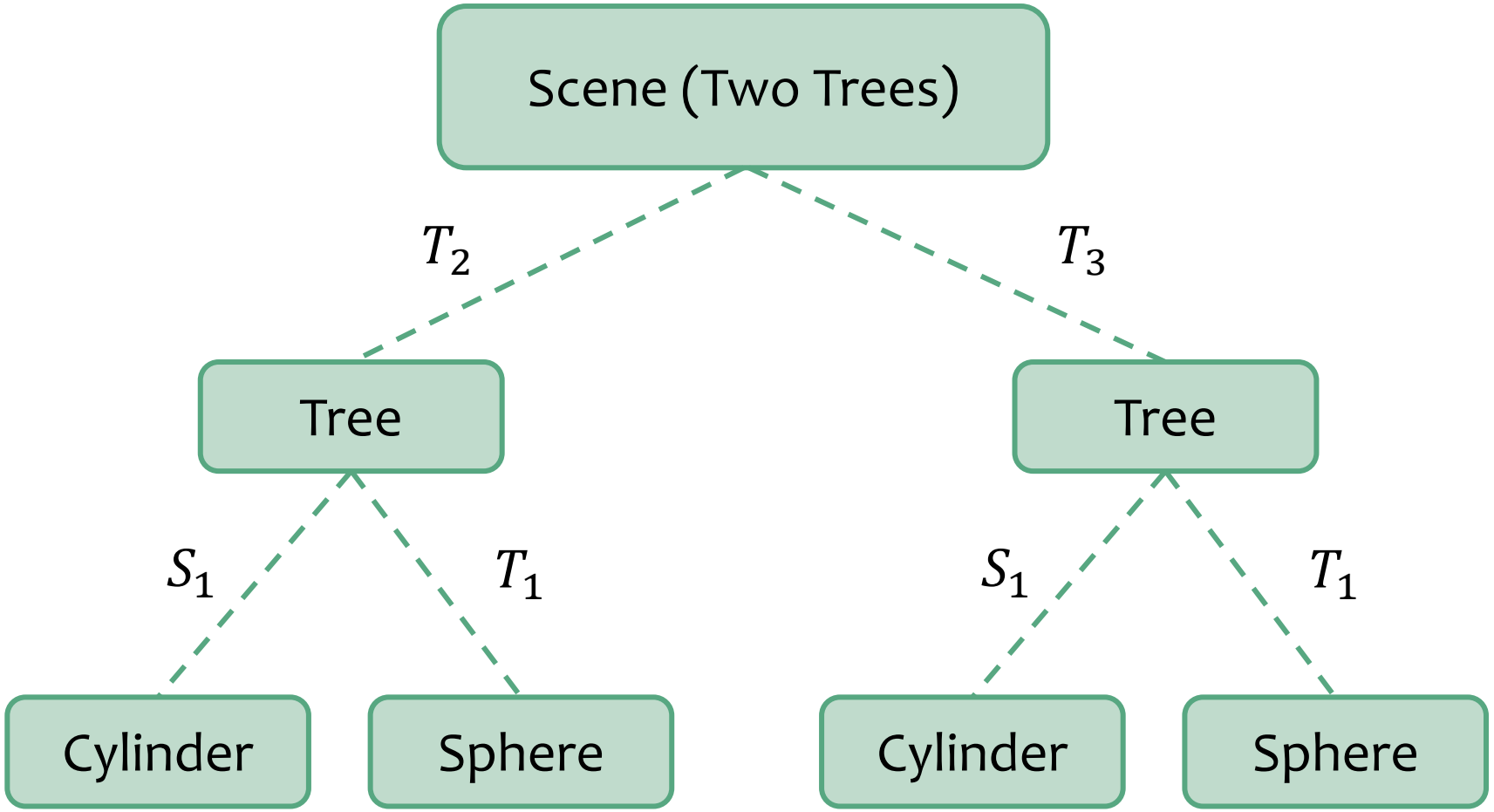


Scene Graphs

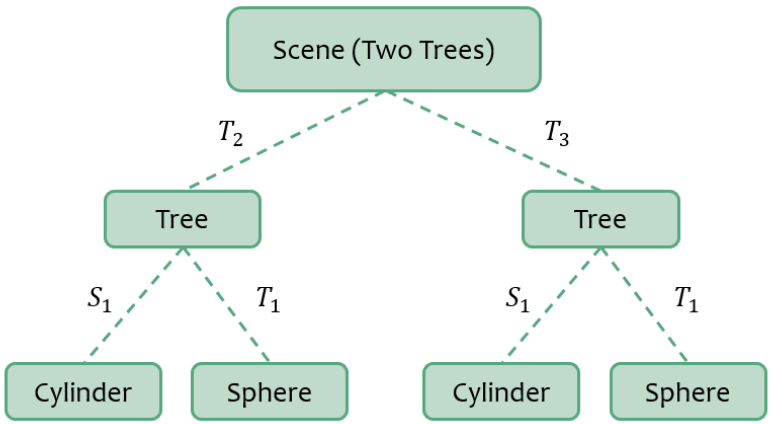
```
tree {
     push
     scale(0.5, 2, 0.5)
     cylinder
     pop
     push
     translate(0, 3, 0)
     sphere
     pop
}
```

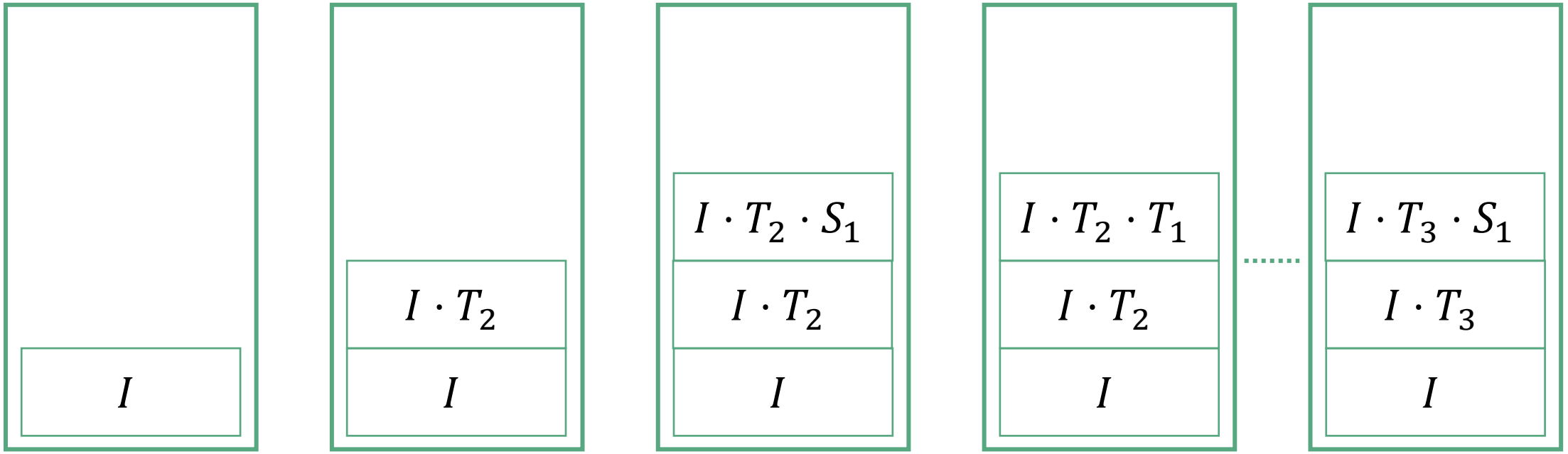


Scene Graphs



Matrix Stack



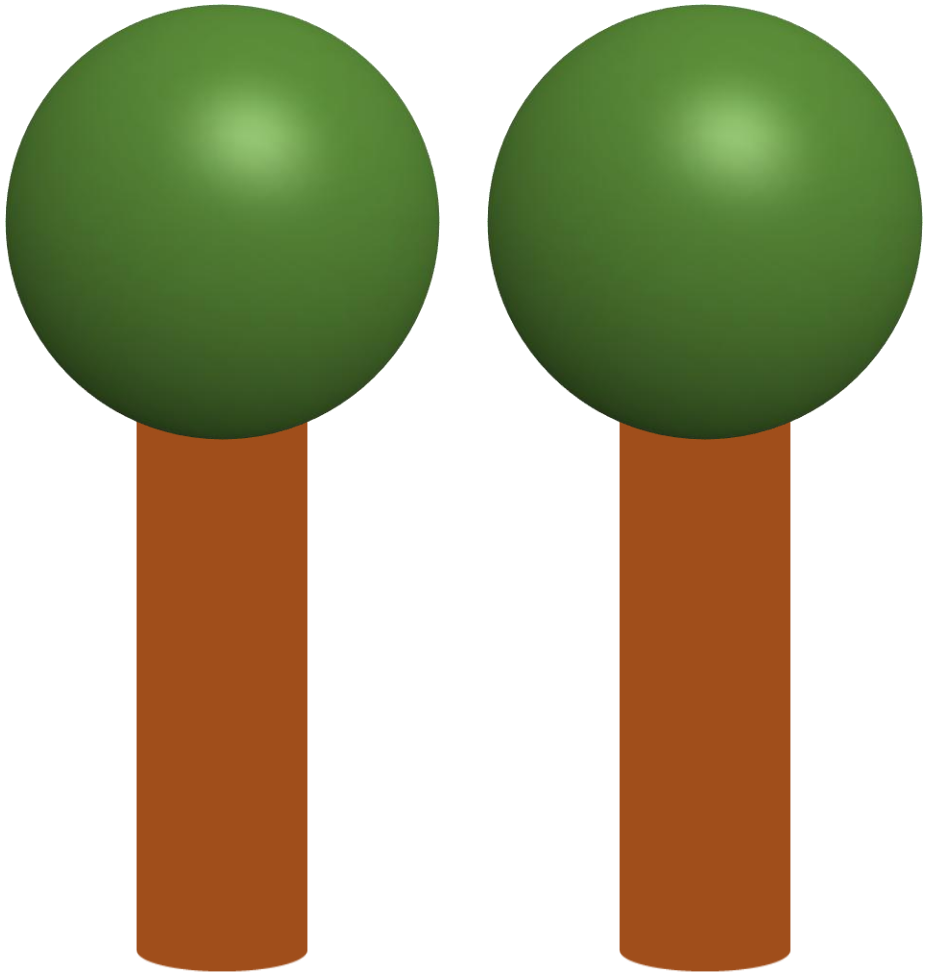


Scene Graphs

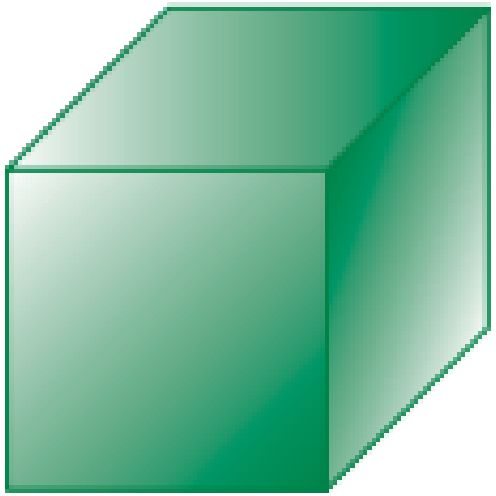
```
two_trees {
      push
      translate(-2, 0, -10)
      tree()
      pop
      push
      translate(-2, 0, -10)
      tree()
      pop
}
```

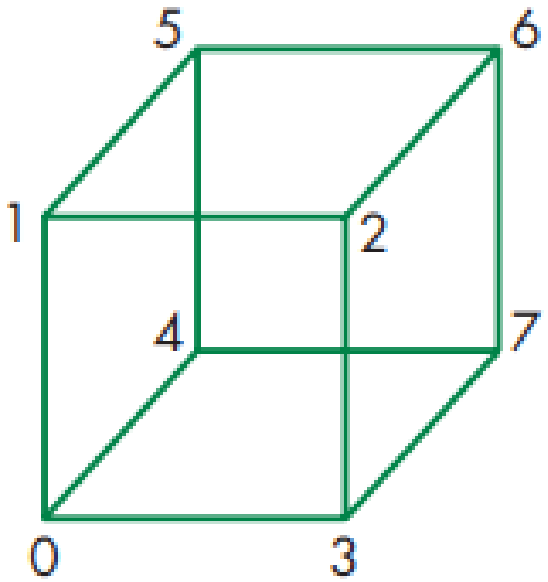
$\rightarrow T_2$

$\rightarrow T_3$



3D Geometry vs Topology





Storing Topology

https://graphics.stanford.edu/data/3Dscanrep/

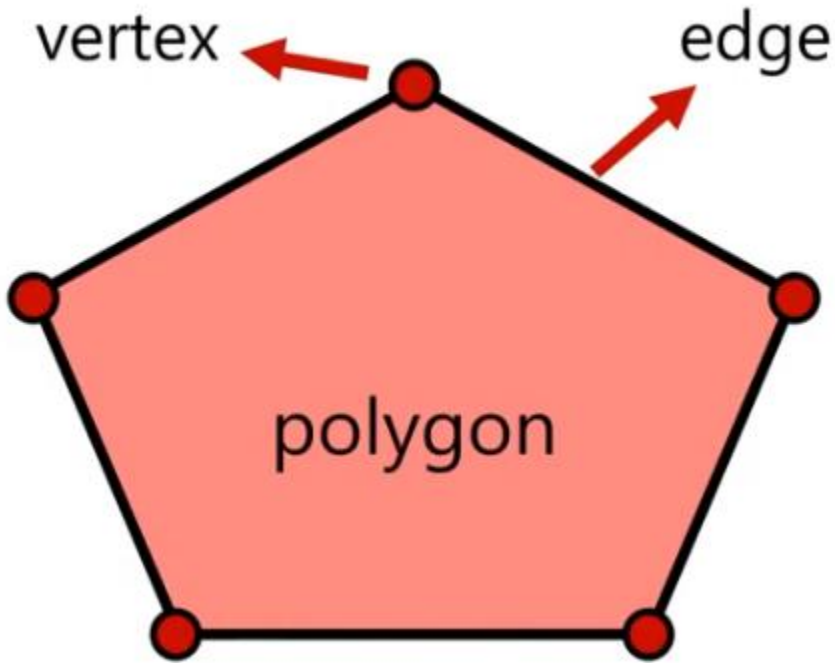
3D Meshes

More about Utah Teapot:
https://youtu.be/TIxt9guMbXo?si=IrZ4zkZ-tusOgCRE

3D Meshes

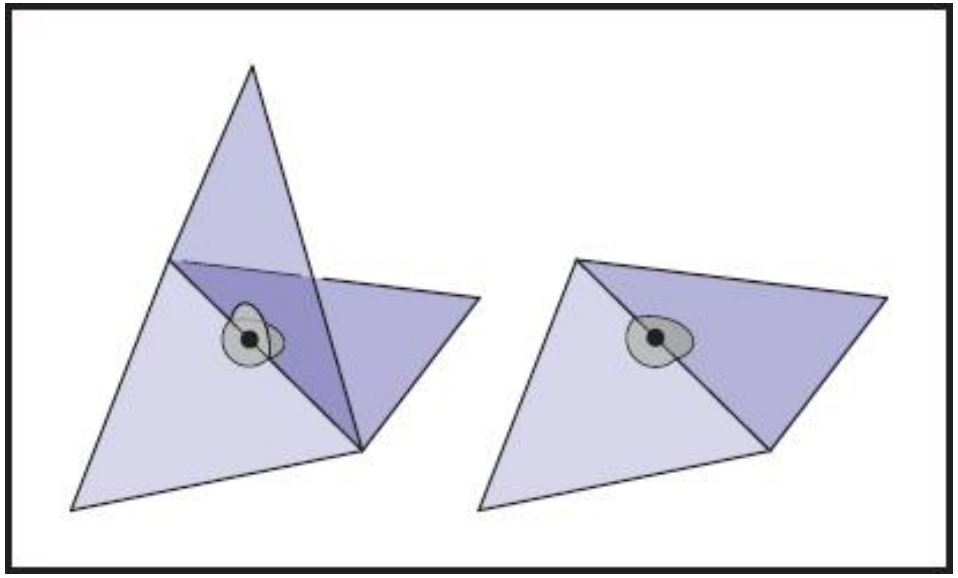
- What are 3D meshes made of?
  - Vertices
  - Edges
  - Faces (or polygons, usually triangles)

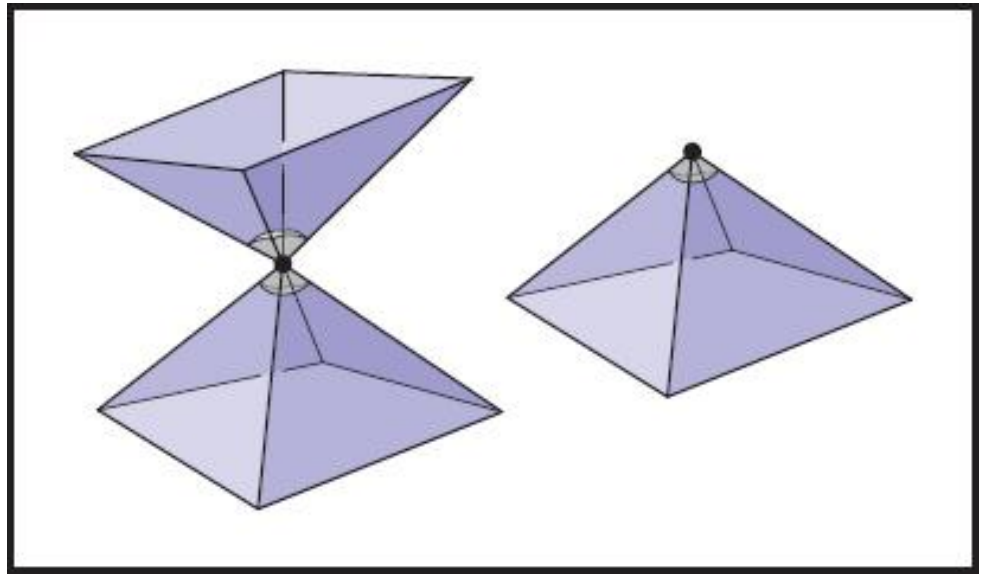
- Meshes represent surfaces



Mesh Topology: Manifold vs Non-manifold

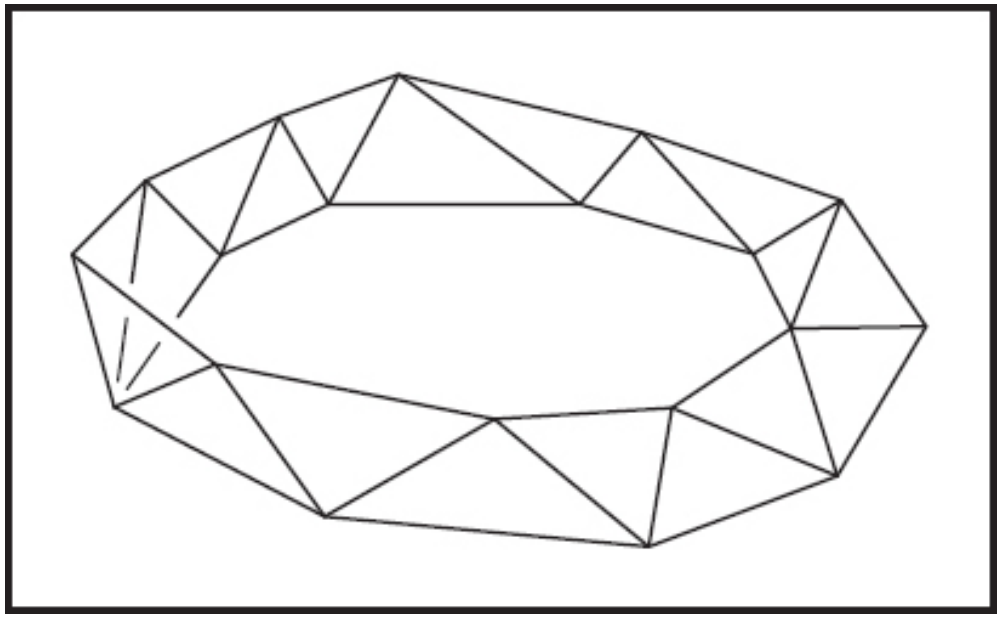
- Edge manifold
- Vertex Manifold

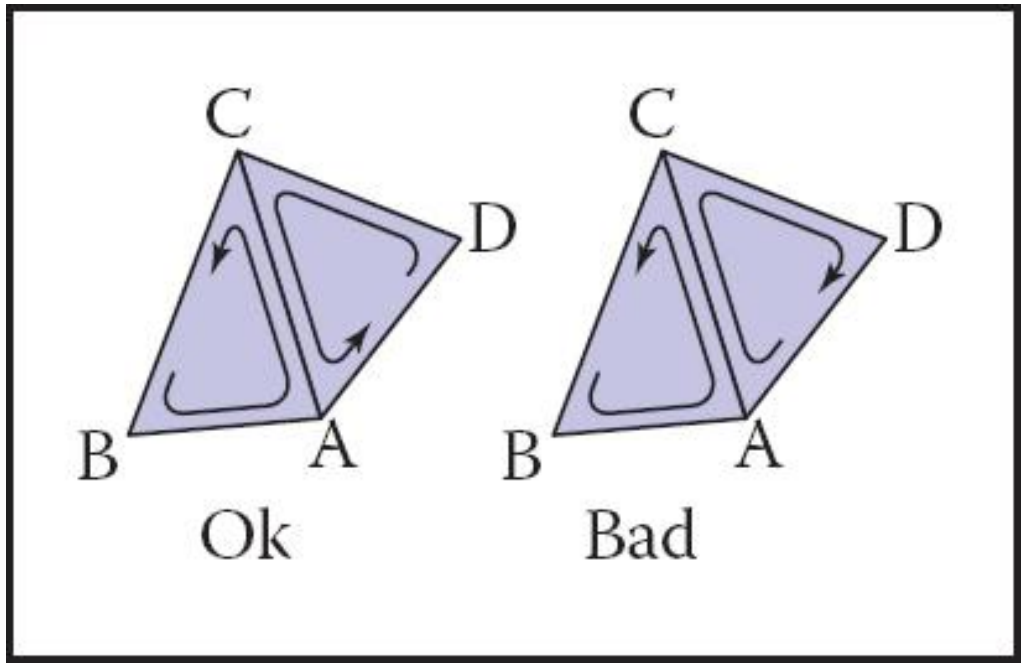




Mesh Topology: Boundary Edges

- Meshes can have **boundaries**
- The can still be **manifold meshes with boundary**



Mesh Topology: Orientation

- We need to distinguish between the “front” and “back” of a face
- Orientation depends on the order of listed vertices
- A mesh is **consistently oriented** if all triangles are defined in the same order